

AI Exploratory Test Report

Target Application: <https://www.calculator.net>

Report Date: March 23, 2026 at 17:27

Tool Version: AI Exploratory Test Engine v2.0

Pages Analyzed: 10

Total Test Cases Generated: 56

Valid Test Cases: 56

Rejected (Hallucinated): 0

Tests Executed: 56

Passed: 56

Failed (Bugs Found): 0

2. Execution Results Summary

56 PASSED

Tests behaved correctly

0 FAILED

Bugs found in application

3. Application Coverage Summary

Metric	Value
Pages Crawled	10
Pages with Interactive Elements	10
Input Fields Identified	75
Buttons Identified	55
Navigation Links	193
Dropdowns/Selects	0
Test Cases Generated	56
Validation Pass Rate	100.0%

4. Batch Generation Strategy

Tests were generated in 6 adversarial batches. Each batch focused on a distinct exploration category to guarantee structural diversity and maximize bug-finding potential.

Batch	Category	Focus
1	Input Torture and Field Poisoning	Edge-case values: long strings, special chars, min-1/max+1, empty required fields.
2	State Transitions and Form Abandonment	Partial fills, navigate-away-return, recalculate after change, state persistence.
3	Boundary Value Analysis	Exact min, max, zero, +1/-1 boundaries across every numeric input field.
4	Multi-Step Navigation Chaos	Fill on page A, jump to B, return to A -- does state survive?
5	Mode Switching and UI State Dances	Toggle modes mid-fill, switch tabs, check value retention after mode change.
6	Error Recovery and Validation Resilience	Submit empty, fix one field, resubmit; clear after success; recovery flow testing.

5. Test Coverage Classification

Category	Count
General Exploratory Tests	22
Boundary Tests	11
Negative Input Tests	9
Zero/Empty Input Tests	7
Validation Enforcement Tests	5
State Persistence Tests	2

Test cases are automatically classified based on their goals and step patterns. Categories are not mutually exclusive; each test is assigned to its primary category.

6. Validated Test Cases

56 test cases passed anti-hallucination validation. All referenced UI elements exist in the application snapshot.

Each test is marked PASSED (green) or FAILED (red) based on execution results.

VT-01 | Break loan calculator with extreme loan amount [PASSED]

Page: /loan-calculator.html

Target: cloanamount

Type: Boundary Tests

1. Navigate to '/loan-calculator.html'
2. Enter '999999999' into 'cloanamount'
3. Click 'Calculate'

Expected: Validation failure

VT-02 | Poison mortgage calculator with special characters [PASSED]

Page: /mortgage-calculator.html

Target: chouseprice

Type: Negative Input Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '!@#\$\$' into 'chouseprice'
3. Click 'Calculate'

*Expected: Error message***VT-03 | Torture auto loan calculator with long string [PASSED]**

Page: /auto-loan-calculator.html

Target: csaleprice

Type: General Exploratory Tests

1. Navigate to '/auto-loan-calculator.html'
2. Enter '01234567890123456789' into 'csaleprice'
3. Click 'Calculate'

*Expected: Validation failure***VT-04 | Contaminate interest calculator with cross-field values [PASSED]**

Page: /interest-calculator.html

Target: cstartingprinciple

Type: Negative Input Tests

1. Navigate to '/interest-calculator.html'
2. Enter '123' into 'cstartingprinciple'
3. Enter '123abc' into 'cannualaddition'
4. Click 'Calculate'

*Expected: Error message***VT-05 | Break payment calculator with empty required field [PASSED]**

Page: /payment-calculator.html

Target: cloanamount

Type: Validation Enforcement Tests

1. Navigate to '/payment-calculator.html'
2. Enter '' into 'cloanamount'
3. Click 'Calculate'

*Expected: Validation failure***VT-06 | Poison retirement calculator with whitespace-only input [PASSED]**

Page: /retirement-calculator.html

Target: cagenow

Type: General Exploratory Tests

1. Navigate to '/retirement-calculator.html'
2. Enter ' ' into 'cagenow'
3. Click 'Calculate'

*Expected: Error message***VT-07 | Torture amortization calculator with negative value [PASSED]**

Page: /amortization-calculator.html

Target: cloanamount

Type: Negative Input Tests

1. Navigate to '/amortization-calculator.html'
2. Enter '-1' into 'cloanamount'
3. Click 'Calculate'

Expected: Validation failure

VT-08 | Contaminate financial calculator with special characters [PASSED]

Page: /financial-calculator.html

Target: calcSearchTerm

Type: Negative Input Tests

1. Navigate to '/financial-calculator.html'
2. Enter '@#\$' into 'calcSearchTerm'
3. Click 'Search'

*Expected: Error message***VT-09 | Break loan calculator with extreme interest rate [PASSED]**

Page: /loan-calculator.html

Target: cinterestrate

Type: Boundary Tests

1. Navigate to '/loan-calculator.html'
2. Enter '999' into 'cinterestrate'
3. Click 'Calculate'

*Expected: Validation failure***VT-10 | Poison mortgage calculator with long string [PASSED]**

Page: /mortgage-calculator.html

Target: cdownpayment

Type: General Exploratory Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '01234567890123456789' into 'cdownpayment'
3. Click 'Calculate'

*Expected: Error message***VT-11 | Test loan calculator with empty loan amount [PASSED]**

Page: /loan-calculator.html

Target: cloanamount

Type: Zero/Empty Input Tests

1. Navigate to '/loan-calculator.html'
2. Enter "" into 'cloanamount'
3. Click 'Calculate'

*Expected: Validation failure for missing loan amount***VT-12 | Test loan calculator with invalid interest rate [PASSED]**

Page: /loan-calculator.html

Target: cinterestrate

Type: Negative Input Tests

1. Navigate to '/loan-calculator.html'
2. Enter 'abc' into 'cinterestrate'
3. Click 'Calculate'

*Expected: Validation failure for invalid interest rate***VT-13 | Test mortgage calculator with empty house price [PASSED]**

Page: /mortgage-calculator.html

Target: chouseprice

Type: Zero/Empty Input Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter "" into 'chouseprice'
3. Click 'Calculate'

Expected: Validation failure for missing house price

VT-14 | Test payment calculator with empty loan term [PASSED]

Page: /payment-calculator.html

Target: cloanterm

Type: Zero/Empty Input Tests

1. Navigate to '/payment-calculator.html'
2. Enter '' into 'cloanterm'
3. Click 'Calculate'

*Expected: Validation failure for missing loan term***VT-15 | Test auto loan calculator with invalid sale price [PASSED]**

Page: /auto-loan-calculator.html

Target: csaleprice

Type: Negative Input Tests

1. Navigate to '/auto-loan-calculator.html'
2. Enter '-1' into 'csaleprice'
3. Click 'Calculate'

*Expected: Validation failure for invalid sale price***VT-16 | Test interest calculator with empty starting principle [PASSED]**

Page: /interest-calculator.html

Target: cstartingprinciple

Type: Zero/Empty Input Tests

1. Navigate to '/interest-calculator.html'
2. Enter '' into 'cstartingprinciple'
3. Click 'Calculate'

*Expected: Validation failure for missing starting principle***VT-17 | Test retirement calculator with empty current age [PASSED]**

Page: /retirement-calculator.html

Target: cagenow

Type: Zero/Empty Input Tests

1. Navigate to '/retirement-calculator.html'
2. Enter '' into 'cagenow'
3. Click 'Calculate'

*Expected: Validation failure for missing current age***VT-18 | Test amortization calculator with empty loan amount [PASSED]**

Page: /amortization-calculator.html

Target: cloanamount

Type: Zero/Empty Input Tests

1. Navigate to '/amortization-calculator.html'
2. Enter '' into 'cloanamount'
3. Click 'Calculate'

*Expected: Validation failure for missing loan amount***VT-19 | Test loan calculator with valid data and then modify loan te [PASSED]**

Page: /loan-calculator.html

Target: cloanamount

Type: General Exploratory Tests

1. Navigate to '/loan-calculator.html'
2. Enter '10000' into 'cloanamount'
3. Enter '5' into 'cloanterm'
4. Click 'Calculate'
5. Enter '10' into 'cloanterm'
6. Click 'Calculate'

Expected: Validation failure for updated loan term

VT-20 | Test minimum allowed value for loan amount [PASSED]

Page: /loan-calculator.html

Target: cloanamount

Type: Boundary Tests

1. Navigate to '/loan-calculator.html'
2. Enter '0' into 'cloanamount'
3. Click 'Calculate'

*Expected: Invalid loan amount***VT-21 | Test maximum allowed value for interest rate [PASSED]**

Page: /loan-calculator.html

Target: cinterestrate

Type: General Exploratory Tests

1. Navigate to '/loan-calculator.html'
2. Enter '100' into 'cinterestrate'
3. Click 'Calculate'

*Expected: Invalid interest rate***VT-22 | Test one below minimum allowed value for loan term [PASSED]**

Page: /loan-calculator.html

Target: cloanterm

Type: Boundary Tests

1. Navigate to '/loan-calculator.html'
2. Enter '-1' into 'cloanterm'
3. Click 'Calculate'

*Expected: Invalid loan term***VT-23 | Test one above maximum allowed value for loan amount [PASSED]**

Page: /loan-calculator.html

Target: cloanamount

Type: Boundary Tests

1. Navigate to '/loan-calculator.html'
2. Enter '9999999999' into 'cloanamount'
3. Click 'Calculate'

*Expected: Invalid loan amount***VT-24 | Test very small decimal value for interest rate [PASSED]**

Page: /loan-calculator.html

Target: cinterestrate

Type: Boundary Tests

1. Navigate to '/loan-calculator.html'
2. Enter '0.0001' into 'cinterestrate'
3. Click 'Calculate'

*Expected: Invalid interest rate***VT-25 | Test zero value for loan term [PASSED]**

Page: /loan-calculator.html

Target: cloanterm

Type: Boundary Tests

1. Navigate to '/loan-calculator.html'
2. Enter '0' into 'cloanterm'
3. Click 'Calculate'

Expected: Invalid loan term

VT-26 | Test minimum allowed value for sale price [PASSED]

Page: /auto-loan-calculator.html

Target: csaleprice

Type: Boundary Tests

1. Navigate to '/auto-loan-calculator.html'
2. Enter '0' into 'csaleprice'
3. Click 'Calculate'

*Expected: Invalid sale price***VT-27 | Test maximum allowed value for annual addition [PASSED]**

Page: /interest-calculator.html

Target: cannualaddition

Type: General Exploratory Tests

1. Navigate to '/interest-calculator.html'
2. Enter '1000000' into 'cannualaddition'
3. Click 'Calculate'

*Expected: Invalid annual addition***VT-28 | Test one below minimum allowed value for monthly addition [PASSED]**

Page: /interest-calculator.html

Target: cmonthlyaddition

Type: Negative Input Tests

1. Navigate to '/interest-calculator.html'
2. Enter '-1' into 'cmonthlyaddition'
3. Click 'Calculate'

*Expected: Invalid monthly addition***VT-29 | Test rapid input submission on loan calculator [PASSED]**

Page: /loan-calculator.html

Target: cloanamount

Type: General Exploratory Tests

1. Navigate to '/loan-calculator.html'
2. Enter '10000' into 'cloanamount'
3. Enter '5' into 'cloanterm'
4. Click 'Calculate'
5. Click 'Calculate' again

*Expected: Validation failure for duplicate submission***VT-30 | Test multi-step form interaction on mortgage calculator [PASSED]**

Page: /mortgage-calculator.html

Target: chouseprice

Type: General Exploratory Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '200000' into 'chouseprice'
3. Enter '20' into 'cdownpayment'
4. Click 'Calculate'
5. Click 'Clear'
6. Enter '300000' into 'chouseprice'
7. Enter '30' into 'cdownpayment'
8. Click 'Calculate'

Expected: Error message for changed input values

VT-31 | Test rapid button clicks on auto loan calculator [PASSED]

Page: /auto-loan-calculator.html

Target: csaleprice

Type: General Exploratory Tests

1. Navigate to '/auto-loan-calculator.html'
2. Enter '20000' into 'csaleprice'
3. Enter '5' into 'cloanterm'
4. Click 'Calculate'
5. Click 'Calculate' again
6. Click 'Calculate' again

*Expected: Validation failure for multiple submissions***VT-32 | Test form interaction with optional fields on payment calcul [PASSED]**

Page: /payment-calculator.html

Target: cloanamount

Type: Zero/Empty Input Tests

1. Navigate to '/payment-calculator.html'
2. Enter '10000' into 'cloanamount'
3. Enter '5' into 'cloanterm'
4. Leave 'cinterestrate' empty
5. Click 'Calculate'

*Expected: Error message for missing required field***VT-33 | Test clear and resubmit on interest calculator [PASSED]**

Page: /interest-calculator.html

Target: cstartingprinciple

Type: General Exploratory Tests

1. Navigate to '/interest-calculator.html'
2. Enter '10000' into 'cstartingprinciple'
3. Enter '5' into 'cannualaddition'
4. Click 'Calculate'
5. Click 'Clear'
6. Enter '20000' into 'cstartingprinciple'
7. Enter '10' into 'cannualaddition'
8. Click 'Calculate'

*Expected: Validation failure for changed input values***VT-34 | Test rapid input submission on retirement calculator [PASSED]**

Page: /retirement-calculator.html

Target: cagenow

Type: General Exploratory Tests

1. Navigate to '/retirement-calculator.html'
2. Enter '30' into 'cagenow'
3. Enter '60' into 'cretireage'
4. Enter '10000' into 'csavingnow'
5. Click 'Calculate'
6. Enter '40' into 'cagenow'
7. Enter '70' into 'cretireage'
8. Enter '20000' into 'csavingnow'
9. Click 'Calculate'

Expected: Error message for changed input values

VT-35 | Test multi-step form interaction on amortization calculator [PASSED]

Page: /amortization-calculator.html

Target: cloanamount

Type: General Exploratory Tests

1. Navigate to '/amortization-calculator.html'
2. Enter '10000' into 'cloanamount'
3. Enter '5' into 'cloanterm'
4. Enter '10' into 'cinterestrate'
5. Click 'Calculate'
6. Click 'Clear'
7. Enter '20000' into 'cloanamount'
8. Enter '10' into 'cloanterm'
9. Enter '20' into 'cinterestrate'
10. Click 'Calculate'

*Expected: Validation failure for changed input values***VT-36 | Test form interaction with required fields on loan calculato [PASSED]**

Page: /loan-calculator.html

Target: cloanamount

Type: Validation Enforcement Tests

1. Navigate to '/loan-calculator.html'
2. Enter '10000' into 'cloanamount'
3. Leave 'cloanterm' empty
4. Click 'Calculate'

*Expected: Error message for missing required field***VT-37 | Test rapid button clicks on payment calculator [PASSED]**

Page: /payment-calculator.html

Target: cloanamount

Type: General Exploratory Tests

1. Navigate to '/payment-calculator.html'
2. Enter '10000' into 'cloanamount'
3. Enter '5' into 'cloanterm'
4. Enter '10' into 'cinterestrate'
5. Click 'Calculate'
6. Click 'Calculate' again
7. Click 'Calculate' again

*Expected: Validation failure for multiple submissions***VT-38 | Break loan calculator by rapid mode switching [PASSED]**

Page: /loan-calculator.html

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/loan-calculator.html'
2. Click 'Calculate'
3. Click 'Clear'
4. Click 'Calculate'

*Expected: Validation failure***VT-39 | Test mortgage calculator with incomplete fields and mode swi [PASSED]**

Page: /mortgage-calculator.html

Target: chouseprice

Type: General Exploratory Tests

1. Navigate to '/mortgage-calculator.html'
2. Click 'Get pre-approval'
3. Enter '100000' into 'chouseprice'
4. Click 'Calculate'

Expected: Error message

VT-40 | Rapidly apply and unapply interest calculator mode [PASSED]

Page: /interest-calculator.html

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/interest-calculator.html'
2. Click 'Calculate'
3. Click 'Clear'
4. Click 'Calculate'
5. Click 'Clear'

Expected: No UI errors

VT-41 | Break auto loan calculator with rapid mode switching [PASSED]

Page: /auto-loan-calculator.html

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/auto-loan-calculator.html'
2. Click 'Calculate'
3. Click 'Search'
4. Click 'Calculate'

Expected: Validation failure

VT-42 | Test payment calculator with mode switch and incomplete fiel [PASSED]

Page: /payment-calculator.html

Target: cloanamount

Type: General Exploratory Tests

1. Navigate to '/payment-calculator.html'
2. Click 'Clear'
3. Enter '10000' into 'cloanamount'
4. Click 'Calculate'

Expected: Error message

VT-43 | Rapidly switch between loan calculator modes [PASSED]

Page: /loan-calculator.html

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/loan-calculator.html'
2. Click 'Calculate'
3. Click 'Clear'
4. Click 'Calculate'
5. Click 'Clear'

Expected: No UI errors

VT-44 | Break amortization calculator with mode switch and incomplet [PASSED]

Page: /amortization-calculator.html

Target: cloanamount

Type: General Exploratory Tests

1. Navigate to '/amortization-calculator.html'
2. Click 'Clear'
3. Enter '10000' into 'cloanamount'
4. Click 'Calculate'

Expected: Error message

VT-45 | Test retirement calculator with rapid mode switching [PASSED]

Page: /retirement-calculator.html

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/retirement-calculator.html'
2. Click 'Calculate'
3. Click 'Clear'
4. Click 'Calculate'

*Expected: Validation failure***VT-46 | Break financial calculator with mode switch and search [PASSED]**

Page: /financial-calculator.html

Target: calcSearchTerm

Type: General Exploratory Tests

1. Navigate to '/financial-calculator.html'
2. Click 'Search'
3. Enter 'loan' into 'calcSearchTerm'
4. Click 'Search'

*Expected: No UI errors***VT-47 | Trigger calculation error and verify error clearance [PASSED]**

Page: /loan-calculator.html

Target: c2loantermmonth

Type: Boundary Tests

1. Navigate to '/loan-calculator.html'
2. Enter '0' into 'c2loantermmonth'
3. Click 'Calculate'
4. Enter '12' into 'c2loantermmonth'
5. Click 'Calculate'

*Expected: Error message is cleared***VT-48 | Submit empty form and verify error message [PASSED]**

Page: /mortgage-calculator.html

Target: N/A

Type: Validation Enforcement Tests

1. Navigate to '/mortgage-calculator.html'
2. Click 'Calculate'

*Expected: Error message is displayed***VT-49 | Submit with invalid value and verify error clearance [PASSED]**

Page: /interest-calculator.html

Target: cstartingprinciple

Type: Negative Input Tests

1. Navigate to '/interest-calculator.html'
2. Enter 'abc' into 'cstartingprinciple'
3. Click 'Calculate'
4. Enter '1000' into 'cstartingprinciple'
5. Click 'Calculate'

*Expected: Error message is cleared***VT-50 | Clear all fields after successful calculation and submit emp [PASSED]**

Page: /payment-calculator.html

Target: cloanamount

Type: State Persistence Tests

1. Navigate to '/payment-calculator.html'
2. Enter '1000' into 'cloanamount'
3. Enter '12' into 'cloanterm'
4. Enter '5' into 'cinterestrate'
5. Click 'Calculate'

6. Click 'Clear'
7. Click 'Calculate'

Expected: Error message is displayed

VT-51 | Enter valid data, click Clear/Reset, and verify error messag [PASSED]

Page: /retirement-calculator.html

Target: cagenow

Type: Validation Enforcement Tests

1. Navigate to '/retirement-calculator.html'
2. Enter '30' into 'cagenow'
3. Enter '60' into 'cretireage'
4. Click 'Clear'
5. Click 'Calculate'

Expected: Error message is displayed

VT-52 | Trigger calculation error and verify error message [PASSED]

Page: /amortization-calculator.html

Target: cloantermmonth

Type: Boundary Tests

1. Navigate to '/amortization-calculator.html'
2. Enter '0' into 'cloantermmonth'
3. Click 'Calculate'

Expected: Error message is displayed

VT-53 | Submit with invalid value and verify error message [PASSED]

Page: /auto-loan-calculator.html

Target: csaleprice

Type: Negative Input Tests

1. Navigate to '/auto-loan-calculator.html'
2. Enter 'abc' into 'csaleprice'
3. Click 'Calculate'

Expected: Error message is displayed

VT-54 | Clear all fields after successful calculation and submit emp [PASSED]

Page: /financial-calculator.html

Target: cloanamount

Type: State Persistence Tests

1. Navigate to '/financial-calculator.html'
2. Click 'Search'
3. Navigate to '/loan-calculator.html'
4. Enter '1000' into 'cloanamount'
5. Enter '12' into 'cloanterm'
6. Enter '5' into 'cinterestrate'
7. Click 'Calculate'
8. Click 'Clear'
9. Click 'Calculate'

Expected: Error message is displayed

VT-55 | Enter valid data, click Clear/Reset, and verify error messag [PASSED]

Page: /mortgage-calculator.html

Target: chouseprice

Type: Validation Enforcement Tests

1. Navigate to '/mortgage-calculator.html'
2. Enter '200000' into 'chouseprice'
3. Enter '20' into 'cdownpayment'
4. Enter '30' into 'cloanterm'
5. Click 'Clear'
6. Click 'Calculate'

Expected: Error message is displayed

VT-56 | Trigger calculation error and verify error clearance [PASSED]

Page: /interest-calculator.html

Target: cyears

Type: Boundary Tests

1. Navigate to '/interest-calculator.html'
2. Enter '0' into 'cyears'
3. Click 'Calculate'
4. Enter '10' into 'cyears'
5. Click 'Calculate'

Expected: Error message is cleared

7. Structural Validation Confirmation

All 56 test cases were validated against the UI exploration snapshot. The validation engine verified that:

- ✓ All referenced UI element labels exist in the application snapshot.
- ✓ All referenced page paths exist in the crawled page inventory.
- ✓ No hallucinated or invented element labels were detected.
- ✓ Each test case has a valid structure: ID, goal, steps (min 2), and expected result.

8. Appendix

Snapshot Statistics

Metric	Value
Snapshot File	ai_exploration_snapshot.json
Snapshot Size	65041 bytes
Total Pages in Snapshot	10
Total Input Fields	75
Total Buttons	55
Total Navigation Links	193

Pipeline Configuration

Parameter	Value
AI Model	llama-3.3-70b-versatile (Grog)
Generation Batches	6
Tests per Batch	~10
Batch Delay	30 seconds
Validation Engine	3-Layer Anti-Hallucination

Step Generator	Deterministic (Template-Based)
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Output Files Reference

File	Path
Raw Test Cases	data/generated_test_cases.json
Validated Test Cases	data/validated_test_cases.json
Rejected Test Cases	data/rejected_test_cases.json
Gauge Specification	specs/ai_exploration.spec
Step Implementations	step_impl/step_implementation.py
This Report	reports/test_report.pdf